CDA 106
Numerical Methods for Data Science

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QR Decomposition

$$A_{m \times k} = \underbrace{Q_{m \times k}}_{\text{orthogonal}} \underbrace{R_{k \times k}}_{\text{upper triangular}}$$

orthogonal

$$QQ^T = I_n$$

$$Q^TQ = I_k$$

Singular Value Decomposition (SVD)

$$A_{m \times n} = U \Sigma V^T$$

Orthogonal

Diagonal(?)

Orthogonal

$\Sigma \rightarrow$ Sigma

σ - sigma

Diagonalizability

A matrix $A_{n \times n}$ is said to be <u>diagonalzable</u>

if $\exists$ a matrix an <u>invertible matrix</u> P such that
(there exists)

Λ : Lambda
λ : lambda.

$$A = P \Lambda P^{-1}$$

eigen vectors of A

eigen values on diagonal

Diagonalization of a Matrix

Assume $\longrightarrow$ $A_{n\times n}$ has n LI eigenvectors

eigenvalues $\lambda_1 \quad \lambda_2 \cdots \lambda_n$

eigenvectors $x_1 \quad x_2 \quad x_n$ —— L.I.

$$A x_i = \lambda_i x_i \quad \forall i$$

$$P = [x_1 | x_2 | \cdots | x_n]$$

Since x_i's are LI, P is invertible $\longrightarrow$ P^{-1} exist.

$$AP = A[x_1 | x_2 | \cdots | x_n] = [Ax_1 | Ax_2 | \cdots | Ax_n]$$

{ Check!! }

$$= [\lambda_1 x_1 | \lambda_2 x_2 | \cdots | \lambda_n x_n] = [x_1 | x_2 | \cdots | x_n] \begin{bmatrix} \lambda_1 & & 0 \\ & \lambda_2 & \\ 0 & & \ddots \; \lambda_n \end{bmatrix} = P\Lambda$$

$$AP = P\Lambda$$

Post multiply by P^{-1} both sides

$$\boxed{A = P\Lambda P^{-1}}$$

diagonal

<u>Imp</u>: P^{-1} <u>exist</u>

<u>Appl.</u> Used in finding A^n !

$A \to$ diagonalizable $\Rightarrow A = P\Lambda P^{-1} \Rightarrow A^2 = (P\Lambda P^{-1})(P\Lambda P^{-1})$

$\Rightarrow A^2 = P\Lambda(P^{-1}P)\Lambda P^{-1} = P\Lambda^2 P^{-1}$ Continuing in same way

$$\boxed{A^n = P\Lambda^n P^{-1}}$$ ✓

<u>Exercise</u> $A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$ then find A^{10}

Q: Can we always diagonalize a matrix?

A: Not possible!

Example $A_{2\times 2} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ $\underbrace{\lambda_1 = 0 \quad \lambda_2 = 0}$ (Repeated)

$(A - 0 \cdot I_2) X = 0$ $x_2 = 0$ & $x_1 =$ arbitrary ($= 1$ for example)

eigen vector

$X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$

$$\boxed{X = \begin{bmatrix} 1 \\ 0 \end{bmatrix}}$$

Here we are unable to get 2 L.I. eigen vectors corr. to $\lambda = 0$

Alg multiplicity $= 2$, Geometric multiplicity $= 1$

Consider Another example.

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 0 \end{bmatrix}$$

$\lambda_1 = 1$, $\lambda_2 = 1$ (Repeated)

(Alg. mult. $= 2$)

$$X_\phi = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

(Geo. mult. $= 1$)

<u>Not diagonalizable</u>

→ <u>Diagonalizable Matrices</u>

eigen values = size of matrix

If all the eigen values are <u>distinct</u> then matrix is diagonalizable.

Example

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

$\lambda_1 = 1$, $\lambda_2 = 2$ (2 eigenvalues)

$$x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \qquad \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$P = [x_1 | x_2] = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \longleftarrow \text{invertible}$$

Therefore

$$A = P \Lambda P^{-1} \quad \text{where} \quad \Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$$

Verify

$$AP = P\Lambda$$

$$\text{LHS} = \begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 4 \\ 0 & 2 \end{bmatrix}$$

$$= P\Lambda \quad \checkmark$$

A real symmetric Matrices

- Real eigenvalues ✓
- Full set of eigenvectors ✓

Therefore, a real symmetric matrix is always diagonalizable.

STRAG Book

- Diagonalizable Matrices

$A = P \Lambda P^{-1}$ Decomposition

- If Matrix $A_{n \times n}$ is real symmetric, then matrix P can be choosen to be orthogonal (Q)

$\boxed{A = Q \Lambda Q^T}$ Decomposition $\left[\begin{array}{l} Q^{-1} = Q^T \\ QQ^T = I \end{array} \right]$

Spectral Theorem

Consider any rectangular matrix

$B_{m \times n}$ then $B^T_{n \times m}$

$B^T B \longrightarrow n \times n$ (square)

$B B^T \longrightarrow m \times m$ (square)

$(B^T B)^T = B^T (B^T)^T = B^T B$ (Symmetric)

$(B B^T)^T = (B^T)^T B^T = B B^T$ (")

Singular Value Decomposition (SVD)

$$A_{m\times n} = \underbrace{U_{m\times m}}_{\text{orthogonal}} \underbrace{\Sigma_{m\times n}}_{\text{diagonal}} \underbrace{V^T_{n\times n}}_{\text{orthogonal}}$$

U: has <u>columns</u> of eigen vectors AA^T [orthogonal, m vectors]

V: has columns of eigen vectors A^TA [orthogonal, n vectors]

Σ: has diagonal entries of square roots of eigenvalues (positive) of AA^T / A^TA

$$A_{m\times n} = \underbrace{\begin{bmatrix} u_1 & u_2 & \cdots & u_m \end{bmatrix}}_{U_{m\times m}} \underbrace{\begin{bmatrix} \sigma_1 & & & 0 \\ & \sigma_2 & & \\ 0 & & \ddots \ \sigma_r & \\ & & & 0 \end{bmatrix}}_{\Sigma_{m\times n}} \underbrace{\begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_n^T \end{bmatrix}}_{V^T_{n\times n}}$$

$$\Sigma_{m\times n} = \begin{bmatrix} D_r & 0 \\ 0 & 0 \end{bmatrix}_{m\times n} \qquad r = \text{rank}(A)$$

$$\sigma_i = \sqrt{\lambda_i} \quad \forall i \ (i = 1 \text{ to } r)$$

Example

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}_{2\times 3}$$

Aim: $A \to SVD$

$r = \text{rank}(A) = \underline{\underline{2}}$

$$A_{2\times 3} = \underbrace{\begin{bmatrix} u_1 & u_2 \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}_{2\times 3}}_{\Sigma} \underbrace{\begin{bmatrix} v_1^T \\ \hline v_2^T \\ \hline v_3^T \end{bmatrix}_{3\times 3}}_{V^T}$$

U → eigenvectors of AA^T

Σ → sq root of eigenvalues

V^T → ??

$\sigma_1 = \sqrt{\lambda_1}, \quad \sigma_2 = \sqrt{\lambda_2}$

Step 1 $AA^T = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} = B$

symm.

$$\lambda^2 - (4)\lambda + 3 = 0$$

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

Step 2 Find eigenvectors of $AA^T = B$

$\lambda_1 = 3$

$$(AA^T - 3I_2)u_1 = 0 \Rightarrow \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0$$

$$u_1^* = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \qquad \|u_1^*\| = \sqrt{2}$$

$$u_1 = \frac{u_1^*}{\|u_1^*\|} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$\lambda_2 = 1$

$$(AA^T - 1I_2)u_2 = 0 \Rightarrow \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = 0 \Rightarrow x_1 - x_2 = 0$$

$$u_2^* = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \qquad \|u_2^*\| = \sqrt{2}$$

$$u_2 = \frac{u_2^*}{\|u_2^*\|} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

One can easily see, for

$$U = [u_1 | u_2] = \begin{bmatrix} \frac{1}{\sqrt{2}} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \quad UU^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

unit unit

and hence U is an orthogonal matrix.

$$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}$$

$$\sigma_2 = \sqrt{\lambda_2} = 1$$

$$\Sigma_{2\times 3} = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}_{2\times 3}$$

Now, we need $V = \left[v_1 \,|\, v_2 \,|\, v_3 \right]_{3\times 3}$

Indirect Calculation

$$v_k = \frac{1}{\sigma_k} A^T u_k \quad , \quad k = 1, 2$$

u_k are the columns of U (unit eigenvectors)

$\underline{\underline{k=1}}$

$$v_1 = \frac{1}{\sigma_1} A^T u_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{6} \\ \sqrt{2/3} \\ -1/\sqrt{6} \end{bmatrix}$$

$k_2 = 2$

$$v_2 = \frac{1}{\sigma_2} A^T u_2 = \frac{1}{1} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

v_3 is required. For that we can find a vector in $\mathbb{R}^3$ such that

$$v_3^T v_1 = 0 \quad , \quad v_3^T v_2 = 0 \quad , \quad v_3^T v_3 = 1$$

$$\boxed{v_3 = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}}$$

eigen values of $A^TA \longrightarrow 3, 1, 0$

unit eigen vector of A^TA corr. to $\lambda = 0$

$\hookrightarrow v_3$ [check !!]

$$V = \begin{bmatrix} -1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \\ \sqrt{2/3} & 0 & 1/\sqrt{3} \\ -1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}$$

$v_1 \quad v_2 \quad v_3$

Finally,

$$A_{2\times 3} = \underbrace{\begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}}_{U} \underbrace{\begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} -1/\sqrt{6} & \sqrt{2/3} & -\frac{1}{\sqrt{6}} \\ -\frac{1}{\sqrt{2}} & 0 & +\frac{1}{\sqrt{2}} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}}_{V^T} \begin{array}{l} \leftarrow v_1^T \\ \leftarrow v_2^T \\ \leftarrow v_3^T \end{array}$$

Therefore, we found the SVD of A.